

Suppose you have to design a machine learning-based solution to recognize Bangla Sign Language (BdSL) words of the BdSLW60 dataset. You can find the dataset description in the link: <https://www.kaggle.com/datasets/hasaniut/bdslw60>. This is a unique dataset that contains 60 daily communication words by 18 signers under the guidance of a professional sign expert. The sign word contains the sequence of hand gestures, including static shape and orientation as well as dynamic movements. Based on the scenario, answer the following questions:

1. Do you think both Convolution Neural Network (CNN) and Recurrent Neural Network (RNN) are required for this sign word recognition task? Justify your answer. [3]
2. Propose a suitable deep learning-based architecture by drawing the block diagram of your proposed solution. [3]
3. Explain how video frames and sign movement information can be processed through your model architecture. [3]
4. Provide relevant mathematical formulations for CNN/RNN operations. [3]
5. Describe how you are going to evaluate your model's performance. Justify your choice on evaluation metric [3]